The Invariance Principle — problem 1.39

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Consider all lattice squares (x, y) with x, y nonnegative integers. Assign to each its lower left corner as a label. We shade the squares $(0, 0)$, $(1, 0)$, $(0, 1)$, $(2, 0)$, $(1, 1)$, $(0, 2)$. (a) There is a chip on each of the six squares (b) There is only one chip on $(0, 0)$.

Step : If (x, y) is occupied, but $(x + 1, y)$ and $(x, y + 1)$ are free, you may remove the chip from (x, y) and place a chip on each of $(x + 1, y)$ and $(x, y + 1)$. The goal is to remove the chips from the shaded squares. Is this possible in the cases (a) or (b) ? (Kontsevich, TT 1981.)